Exponentials & Logarithms

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The Natural Exponential & Logarithm**

An interval of width 200 containing a root must be reduced to width at most 1 using repeated bisection. Find the minimum number of bisections required.

Halving repeatedly, after n bisections the width is $200/2^n$. The smallest n with $200/2^n \le 1$ is $n = 8$ (width becomes $\frac{25}{32}$). Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

2 **[Foundational]** • **Target: Exponential Equations with Different Bases**

Evaluate $\int_1^2 (2x+5)^1\,dx$.

$\int (2x+5)^1 dx = \frac{(2x+5)^2}{2(2)}$. At $x = 2$: base $= 9$, giving 81. At $x = 1$: base $= 7$, giving 49. Result: $\frac{81-49}{4} = 8$. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

3 **[Foundational]** • **Target: Transforming to Linear Form ($y = Ax^n$ and $y = Ab^x$)**

Evaluate $\int_0^{\ln(8)/4} 5e^{4x}\,dx$.

$\int 5e^{4x}\,dx = \frac{5}{4}e^{4x}$. At $x = \ln(8)/4$: $e^{4x} = 8$. At $x = 0$: $e^{4x} = 1$. Result: $\frac{5}{4}(8-1) = \frac{35}{4}$. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

4 **[Foundational]** • **Target: Exponential Growth and Decay Modeling**

Evaluate $\int_0^{\ln(4)/2} e^{2x}\,dx$.

$\int e^{2x}\,dx = \frac{1}{2}e^{2x}$. At $x = \ln(4)/2$: $e^{2x} = 4$. At $x = 0$: $e^{2x} = 1$. Result: $\frac{1}{2}(4-1) = \frac{3}{2}$. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

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5 [Foundational] • Target: The Natural Exponential & Logarithm

Solve $|4x + 12| > 6$. What is the smaller boundary value separating the two regions?

$4x + 12 > 6$ or $4x + 12 < -6$, giving boundary values $x = -\frac{3}{2}$ and $x = -\frac{9}{2}$. Smaller boundary: $x = -\frac{9}{2}$. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

6 [Foundational] • Target: Exponential Equations with Different Bases

Find $\int \frac{2x}{(x+2)(x-1)}\,dx$.

Partial fractions give $\int \frac{2x}{(x+2)(x-1)}\,dx = \frac{2\log(x-1)}{3} + \frac{4\log(x+2)}{3} + C$. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

7 [Foundational] • Target: Transforming to Linear Form ($y = Ax^n$ and $y = Ab^x$)

Evaluate $\int_0^5 (x-12)^5\,dx$.

$\int (x-12)^5 dx = \frac{(x-12)^6}{(6)}$. At $x = 5$: base $= -7$, giving 117649. At $x = 0$: base $= -12$, giving 2985984. Result: $\frac{117649 - 2985984}{6} = -\frac{2868335}{6}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Exponential Growth and Decay Modeling

Evaluate $\int_0^4 (4x-5)^4\,dx$.

$\int (4x-5)^4 dx = \frac{(4x-5)^5}{4(5)}$. At $x = 4$: base $= 11$, giving 161051. At $x = 0$: base $= -5$, giving -3125. Result: $\frac{161051 + 3125}{20} = \frac{41044}{5}$.

9 [Intermediate] • Target: The Natural Exponential & Logarithm

Evaluate $\int_{-1}^0 (3x-8)^1\,dx$.

$\int (3x-8)^1 dx = \frac{(3x-8)^2}{3(2)}$. At $x = 0$: base $= -8$, giving 64. At $x = -1$: base $= -11$, giving 121. Result: $\frac{64 - 121}{6} = -\frac{19}{2}$.

10 [Intermediate] • Target: Exponential Equations with Different Bases

Evaluate $\int_{-2}^1 (5x+8)^4\,dx$.

$\int (5x+8)^4 dx = \frac{(5x+8)^5}{5(5)}$. At $x = 1$: base $= 13$, giving 371293. At $x = -2$: base $= -2$, giving -32. Result: $\frac{371293 + 32}{25} = 14853$.

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11 [Intermediate] • Target: Transforming to Linear Form ($y = Ax^n$ and $y = Ab^x$)

Solve $e^{2x} - 23e^x + 130 = 0$ for x. Give the larger solution in exact form.

Final Answer: $\ln 13$. $(e^x - 10)(e^x - 13) = 0 \Rightarrow e^x = 10$ or $e^x = 13 \Rightarrow x = \ln 10$ or $x = \ln 13$. Larger: $x = \ln 13$.

12 [Intermediate] • Target: Exponential Growth and Decay Modeling

Solve $e^{2x} - 21e^x + 110 = 0$ for x. Give the larger solution in exact form.

Final Answer: $\ln 11$. $(e^x - 10)(e^x - 11) = 0 \Rightarrow e^x = 10$ or $e^x = 11 \Rightarrow x = \ln 10$ or $x = \ln 11$. Larger: $x = \ln 11$.

13 [Intermediate] • Target: The Natural Exponential & Logarithm

Solve $e^{2x} - 28e^x + 187 = 0$ for x. Give the larger solution in exact form.

Final Answer: $\ln 17$. $(e^x - 11)(e^x - 17) = 0 \Rightarrow e^x = 11$ or $e^x = 17 \Rightarrow x = \ln 11$ or $x = \ln 17$. Larger: $x = \ln 17$.

14 [Intermediate] • Target: Exponential Equations with Different Bases

Solve $e^{2x} - 22e^x + 120 = 0$ for x. Give the larger solution in exact form.

Final Answer: $\ln 12$. $(e^x - 10)(e^x - 12) = 0 \Rightarrow e^x = 10$ or $e^x = 12 \Rightarrow x = \ln 10$ or $x = \ln 12$. Larger: $x = \ln 12$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Transforming to Linear Form ($y = Ax^n$ and $y = Ab^x$) • Cross-Lesson: 3.01 Advanced Trigonometry

Solve $e^{2x} - 39e^x + 360 = 0$ for x. Give the larger solution in exact form.

Final Answer: $\ln 24$. $(e^x - 15)(e^x - 24) = 0 \Rightarrow e^x = 15$ or $e^x = 24 \Rightarrow x = \ln 15$ or $x = \ln 24$. Larger: $x = \ln 24$.

16 [Challenging] • Target: Exponential Growth and Decay Modeling • Cross-Lesson: 4.01 Differentiation

Solve $e^{2x} - 22e^x + 112 = 0$ for x. Give the larger solution in exact form.

Final Answer: $\ln 14$. $(e^x - 8)(e^x - 14) = 0 \Rightarrow e^x = 8$ or $e^x = 14 \Rightarrow x = \ln 8$ or $x = \ln 14$. Larger: $x = \ln 14$.

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17 [Challenging] • Target: The Natural Exponential & Logarithm • Cross-Lesson: 5.01 Integration

Solve $e^{2x} - 13e^x + 30 = 0$ for x. Give the larger solution in exact form.

Final Answer: $\ln 10$. $(e^x - 3)(e^x - 10) = 0 \Rightarrow e^x = 3$ or $e^x = 10 \Rightarrow x = \ln 3$ or $x = \ln 10$. Larger: $x = \ln 10$.

18 [Challenging] • Target: Exponential Equations with Different Bases • Cross-Lesson: 6.01 Sequences & Series

Solve $e^{2x} - 24e^x + 108 = 0$ for x. Give the larger solution in exact form.

Final Answer: $\ln 18$. $(e^x - 6)(e^x - 18) = 0 \Rightarrow e^x = 6$ or $e^x = 18 \Rightarrow x = \ln 6$ or $x = \ln 18$. Larger: $x = \ln 18$.

19 [Challenging] • Target: Transforming to Linear Form ($y = Ax^n$ and $y = Ab^x$) • Cross-Lesson: 7.01 Vectors

Solve $e^{2x} - 17e^x + 42 = 0$ for x. Give the larger solution in exact form.

Final Answer: $\ln 14$. $(e^x - 3)(e^x - 14) = 0 \Rightarrow e^x = 3$ or $e^x = 14 \Rightarrow x = \ln 3$ or $x = \ln 14$. Larger: $x = \ln 14$.

20 [Challenging] • Target: Exponential Growth and Decay Modeling • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

Solve $e^{2x} - 19e^x + 48 = 0$ for x. Give the larger solution in exact form.

Final Answer: $\ln 16$. $(e^x - 3)(e^x - 16) = 0 \Rightarrow e^x = 3$ or $e^x = 16 \Rightarrow x = \ln 3$ or $x = \ln 16$. Larger: $x = \ln 16$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The Natural Exponential & Logarithm

Solve the equation $e^{2x-1} = 5$, giving your answer in exact form.

a. $x = \frac{1 + \ln 5}{2}$ ✓

b. $x = \frac{\ln 5 - 1}{2}$

c. $x = \ln\left(\frac{5}{2}\right) + 1$

d. $x = \frac{\ln 6}{2}$

Take natural logarithms of both sides: $2x - 1 = \ln 5 \Rightarrow 2x = 1 + \ln 5 \Rightarrow x = \frac{1 + \ln 5}{2}$.

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22 [Foundational] • Target: The Natural Exponential & Logarithm

Solve $\ln(3x+2) - \ln(x) = \ln 4$ for $x > 0$.

a. $x = 2$ ✓

b. $x = 1$

c. $x = 4$

d. $x = 1/2$

$\ln\left(\frac{3x+2}{x}\right) = \ln 4 \Rightarrow \frac{3x+2}{x} = 4 \Rightarrow 3x+2 = 4x \Rightarrow x = 2$.

23 [Foundational] • Target: The Natural Exponential & Logarithm

Simplify the expression $e^{3\ln x} + \ln(e^{2x})$ for $x > 0$.

a. $x^3 + 2x$ ✓

b. $3x + 2x = 5x$

c. $x^3 + e^{2x}$

d. $3x^2 + 2$

$e^{3\ln x} = e^{\ln(x^3)} = x^3$. $\ln(e^{2x}) = 2x$. Sum is $x^3 + 2x$.

24 [Foundational] • Target: Exponential Equations with Different Bases

Solve the equation $2^{x+1} = 3^x$, giving x in terms of natural logarithms.

a. $x = \frac{\ln 2}{\ln 3 - \ln 2}$ ✓

b. $x = \frac{\ln 3}{\ln 3 - \ln 2}$

c. $x = \frac{\ln 2}{\ln 3 + \ln 2}$

d. $x = \frac{1}{\ln(3/2)}$

$\ln(2^{x+1}) = \ln(3^x) \Rightarrow (x+1)\ln 2 = x\ln 3 \Rightarrow x\ln 2 + \ln 2 = x\ln 3 \Rightarrow x(\ln 3 - \ln 2) = \ln 2 \Rightarrow x = \frac{\ln 2}{\ln 3 - \ln 2}$.

25 [Foundational] • Target: Exponential Equations with Different Bases

Solve the equation $e^x - 6e^{-x} = 1$ for real x.

a. $x = \ln 3$ ✓

b. $x = \ln 3$ or $x = \ln(-2)$

c. $x = \ln 2$

d. $x = 3$

Multiply by e^x: $(e^x)^2 - e^x - 6 = 0 \Rightarrow (e^x - 3)(e^x + 2) = 0$. Since $e^x > 0$, $e^x = 3 \Rightarrow x = \ln 3$.

26 [Foundational] • Target: Exponential Equations with Different Bases

Solve $5^{2x-1} = 7$, giving your answer correct to 3 significant figures.

a. 1.10 ✓

b. 0.852

c. 1.21

d. 0.945

$(2x-1)\ln 5 = \ln 7 \Rightarrow 2x - 1 = \frac{\ln 7}{\ln 5} = \frac{1.9459}{1.6094} \approx 1.209 \Rightarrow 2x \approx 2.209 \Rightarrow x \approx 1.10$.

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27 [Foundational] • Target: Transforming to Linear Form ($y = Ax^n$ and $y = Ab^x$)

The variables x and y satisfy $y = ab^x$. When $\ln y$ is plotted against x, a straight line is obtained. What are the gradient and vertical intercept?

a. **Gradient** $= \ln b$**, Intercept** $= \ln a$ ✓

b. Gradient $= b$, Intercept $= a$

c. Gradient $= \ln a$, Intercept $= \ln b$

d. Gradient $= a$, Intercept $= \ln b$

Take logarithms: $\ln y = \ln(a \cdot b^x) = \ln a + x \ln b = (\ln b)x + \ln a$. This matches $Y = mX + c$ with gradient $\ln b$ and intercept $\ln a$.

28 [Foundational] • Target: Transforming to Linear Form ($y = Ax^n$ and $y = Ab^x$)

The variables x and y satisfy $y = kx^n$. The line of $\ln y$ against $\ln x$ has gradient 2.5 and intercept $\ln y = 1.386$. Find k and n.

a. $k = 4, n = 2.5$ ✓

b. $k = 2.5, n = 4$

c. $k = 1.386, n = 2.5$

d. $k = e^{2.5}, n = 1.386$

$\ln y = n \ln x + \ln k$. Gradient $n = 2.5$. Intercept $\ln k = 1.386 \Rightarrow k = e^{1.386} = 4$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Transforming to Linear Form ($y = Ax^n$ and $y = Ab^x$)

Points $(2, 3.2)$ and $(5, 5.6)$ lie on the straight line of $\ln y$ against x. Find the value of b if $y = ab^x$.

a. $b = e^{0.8} \approx 2.226$ ✓

b. $b = 0.8$

c. $b = 2.4$

d. $b = e^{1.6} \approx 4.95$

Gradient $m = \ln b = \frac{5.6 - 3.2}{5 - 2} = \frac{2.4}{3} = 0.8$. Thus $b = e^{0.8} \approx 2.226$.

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30 [Intermediate] • Target: Exponential Growth and Decay Modeling

The mass of a radioactive substance decays according to $M(t) = 80e^{-0.05t}$, where t is in years. Find the half-life of the substance in years.

a. ≈ 13.9 years ✓

b. 20.0 years

c. 10.0 years

d. 15.5 years

$M(t) = 40 \Rightarrow 80e^{-0.05t} = 40 \Rightarrow e^{-0.05t} = 0.5 \Rightarrow -0.05t = -\ln 2 \Rightarrow t = \frac{\ln 2}{0.05} = 20\ln 2 \approx 13.86$ years.

31 [Intermediate] • Target: Exponential Growth and Decay Modeling

A bacterial culture grows according to $N(t) = 500e^{kt}$. If $N(3) = 2000$, find the exact value of k.

a. $k = \frac{\ln 4}{3}$ ✓

b. $k = \frac{\ln 2}{3}$

c. $k = \frac{4}{3}$

d. $k = \ln\left(\frac{4}{3}\right)$

$500e^{3k} = 2000 \Rightarrow e^{3k} = 4 \Rightarrow 3k = \ln 4 \Rightarrow k = \frac{\ln 4}{3}$.

32 [Intermediate] • Target: Exponential Growth and Decay Modeling

A hot beverage cools according to Newton's law of cooling: $T(t) = 20 + 70e^{-0.08t}$, where T is temperature in $^\circ$C and t is time in minutes. How long does it take to cool to $45\,^\circ$C?

a. ≈ 12.9 min ✓

b. 15.2 min

c. 10.5 min

d. 18.0 min

$45 = 20 + 70e^{-0.08t} \Rightarrow 25 = 70e^{-0.08t} \Rightarrow e^{-0.08t} = 25/70 = 5/14 \Rightarrow -0.08t = \ln(5/14) \approx -1.0296 \Rightarrow t \approx 1.0296/0.08 \approx 12.87$ min.

33 [Intermediate] • Target: The Natural Exponential & Logarithm

Solve $e^{2x} - 14e^x + 33 = 0$ for x. Give the larger solution in exact form.

a. $\ln 11$ ✓

b. $\ln 14$

c. $\ln 33$

d. $\ln 3$

$(e^x - 3)(e^x - 11) = 0 \Rightarrow e^x = 3$ or $e^x = 11 \Rightarrow x = \ln 3$ or $x = \ln 11$. Larger: $x = \ln 11$.

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34 [Intermediate] • Target: Exponential Equations with Different Bases

Solve $e^{2x} - 25e^x + 144 = 0$ for x. Give the larger solution in exact form.

a. $\ln 16$ ✓

b. $\ln 9$

c. $\ln 25$

d. $\ln 144$

$(e^x - 9)(e^x - 16) = 0 \Rightarrow e^x = 9$ or $e^x = 16 \Rightarrow x = \ln 9$ or $x = \ln 16$. Larger: $x = \ln 16$.

35 [Intermediate] • Target: Transforming to Linear Form ($y = Ax^n$ and $y = Ab^x$)

Solve $e^{2x} - 26e^x + 120 = 0$ for x. Give the larger solution in exact form.

a. $\ln 6$

b. $\ln 120$

c. $\ln 20$ ✓

d. $\ln 26$

$(e^x - 6)(e^x - 20) = 0 \Rightarrow e^x = 6$ or $e^x = 20 \Rightarrow x = \ln 6$ or $x = \ln 20$. Larger: $x = \ln 20$.

36 [Intermediate] • Target: Exponential Growth and Decay Modeling

Solve $e^{2x} - 15e^x + 44 = 0$ for x. Give the larger solution in exact form.

a. $\ln 15$

b. $\ln 11$ ✓

c. $\ln 4$

d. $\ln 44$

$(e^x - 4)(e^x - 11) = 0 \Rightarrow e^x = 4$ or $e^x = 11 \Rightarrow x = \ln 4$ or $x = \ln 11$. Larger: $x = \ln 11$.

37 [Intermediate] • Target: The Natural Exponential & Logarithm

Solve $e^{2x} - 27e^x + 140 = 0$ for x. Give the larger solution in exact form.

a. $\ln 20$ ✓

b. $\ln 27$

c. $\ln 140$

d. $\ln 7$

$(e^x - 7)(e^x - 20) = 0 \Rightarrow e^x = 7$ or $e^x = 20 \Rightarrow x = \ln 7$ or $x = \ln 20$. Larger: $x = \ln 20$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Exponential Equations with Different Bases • Cross-Lesson: 6.01 Sequences & Series

Solve $e^{2x} - 28e^x + 52 = 0$ for x. Give the larger solution in exact form.

a. $\ln 52$

b. $\ln 26$ ✓

c. $\ln 2$

d. $\ln 28$

$(e^x - 2)(e^x - 26) = 0 \Rightarrow e^x = 2$ or $e^x = 26 \Rightarrow x = \ln 2$ or $x = \ln 26$. Larger: $x = \ln 26$.

39 [Challenging] • Target: Transforming to Linear Form ($y = Ax^n$ and $y = Ab^x$) • Cross-Lesson: 7.01 Vectors

Solve $e^{2x} - 23e^x + 42 = 0$ for x. Give the larger solution in exact form.

a. $\ln 21$ ✓

b. $\ln 42$

c. $\ln 2$

d. $\ln 23$

$(e^x - 2)(e^x - 21) = 0 \Rightarrow e^x = 2$ or $e^x = 21 \Rightarrow x = \ln 2$ or $x = \ln 21$. Larger: $x = \ln 21$.

40 [Challenging] • Target: Exponential Growth and Decay Modeling • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

Solve $e^{2x} - 20e^x + 91 = 0$ for x. Give the larger solution in exact form.

a. $\ln 91$

b. $\ln 20$

c. $\ln 13$ ✓

d. $\ln 7$

$(e^x - 7)(e^x - 13) = 0 \Rightarrow e^x = 7$ or $e^x = 13 \Rightarrow x = \ln 7$ or $x = \ln 13$. Larger: $x = \ln 13$.

41 [Challenging] • Target: The Natural Exponential & Logarithm • Cross-Lesson: 9.01 Complex Numbers

Solve $e^{2x} - 7e^x + 10 = 0$ for x. Give the larger solution in exact form.

a. $\ln 2$

b. $\ln 10$

c. $\ln 7$

d. $\ln 5$ ✓

$(e^x - 2)(e^x - 5) = 0 \Rightarrow e^x = 2$ or $e^x = 5 \Rightarrow x = \ln 2$ or $x = \ln 5$. Larger: $x = \ln 5$.

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42 [Challenging] • Target: Exponential Equations with Different Bases • Cross-Lesson: 1.01 Functions: Modulus & Inverse

Solve $e^{2x} - 7e^x + 12 = 0$ for x. Give the larger solution in exact form.

a. $\ln 12$

b. $\ln 3$

c. $\ln 4$ ✓

d. $\ln 7$

$(e^x - 3)(e^x - 4) = 0 \Rightarrow e^x = 3$ or $e^x = 4 \Rightarrow x = \ln 3$ or $x = \ln 4$. Larger: $x = \ln 4$.

43 [Challenging] • Target: Transforming to Linear Form ($y = Ax^n$ and $y = Ab^x$) • Cross-Lesson: 10.01 Numerical Methods

Solve $e^{2x} - 13e^x + 42 = 0$ for x. Give the larger solution in exact form.

a. $\ln 6$

b. $\ln 42$

c. $\ln 7$ ✓

d. $\ln 13$

$(e^x - 6)(e^x - 7) = 0 \Rightarrow e^x = 6$ or $e^x = 7 \Rightarrow x = \ln 6$ or $x = \ln 7$. Larger: $x = \ln 7$.

44 [Challenging] • Target: Exponential Growth and Decay Modeling • Cross-Lesson: 11.01 Differential Equations

Find the set of all real solutions to the equation $\log_2(x-1) - \log_4(x+14) = 1$

a. $x = 11$ **only** ✓

b. $x = 11$ and $x = -5$

c. $x = 5$ only

d. $x = 16$ only

To solve the equation, we first express all logarithms in the same base using the change of base formula: $\log_4(x+14) = \frac{\log_2(x+14)}{\log_2 4} = \frac{\log_2(x+14)}{2} = \frac{1}{2}\log_2(x+14)$ Substitute this back into the original equation: $\log_2(x-1) - \frac{1}{2}\log_2(x+14) = 1$ Multiply through by 2: $2\log_2(x-1) - \log_2(x+14) = 2$ Apply the power law and the quotient law of logarithms.

45 [Challenging] • Target: The Natural Exponential & Logarithm • Cross-Lesson: 12.01 Poisson Distribution

Determine the complete set of real values of the constant k for which the equation $e^x - kx^2 = 0$ has exactly three distinct real roots.

a. $k > \frac{e^2}{4}$ ✓

b. $k \geq \frac{e^2}{4}$

c. $0 < k < \frac{e^2}{4}$

d. $k > \frac{e}{2}$

First, note that $x = 0$ is not a solution to $e^x - kx^2 = 0$ since $e^0 = 1 eq 0$. Therefore, we can rewrite the equation as: $k = \frac{e^x}{x^2}$ Let $f(x) = \frac{e^x}{x^2}$ defined for all $x eq 0$. The number of distinct real roots corresponds to the number of points of intersection between the curve $y = f(x)$ and the horizontal line $y = k$. Differentiating $f(x)$ using the quotient rule: $f'(x) = \frac{e^x \cdot x^2 - e^x \cdot 2x}{x^4} = \frac{e^x x(x-2)}{x^4} = \frac{e^x(x-2)}{x^3}$ Setting.

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